

Question ID: 468613c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A triangle has a base length of **56** centimeters and a height of **112** centimeters. What is the area, in square centimeters, of the triangle?

Answer

- A. **168**
- B. **1,568**
- C. **3,136**
- D. **6,272**

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the base length and h is the height of the triangle. It's given that a triangle has a base length of **56** centimeters and a height of **112** centimeters. Substituting **56** for b and **112** for h in the formula $A = \frac{1}{2}bh$ yields $A = (\frac{1}{2})(56)(112)$, or $A = 3,136$. Therefore, the area, in square centimeters, of the triangle is **3,136**.

Choice A is incorrect. This is the value of $56 + 112$, not $(\frac{1}{2})(56)(112)$.

Choice B is incorrect. This is the value of $(\frac{1}{4})(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Choice D is incorrect. This is the value of $(56)(112)$, not $(\frac{1}{2})(56)(112)$.